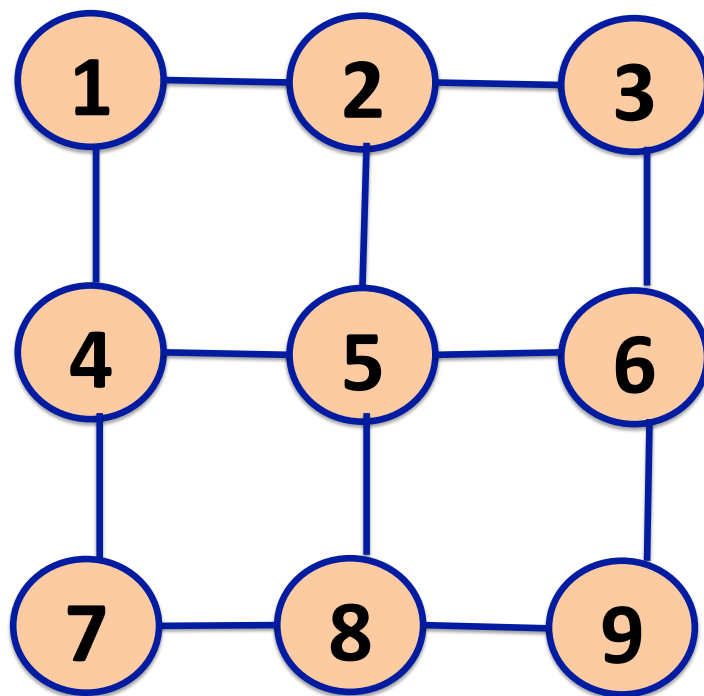




Think about the worst case delay from a possible location of the controller to the nodes in the rest of the network

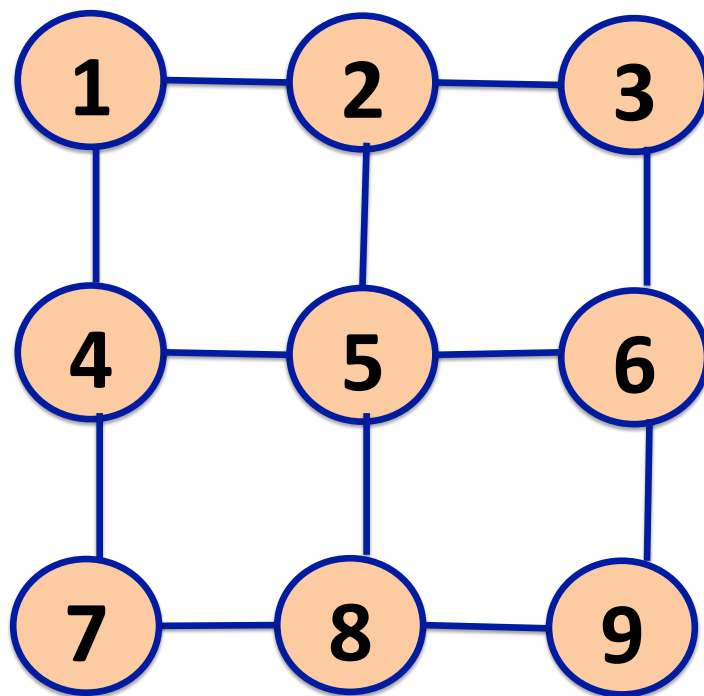


Controller Node 5:

Delay 5-2: 1

Delay 5-1: 2

Maximum delay between the controller  
and any packet switch: 2



Controller Node 1:

Delay 1-5: 2

Delay 1-6: 3

Delay 1-9: 4

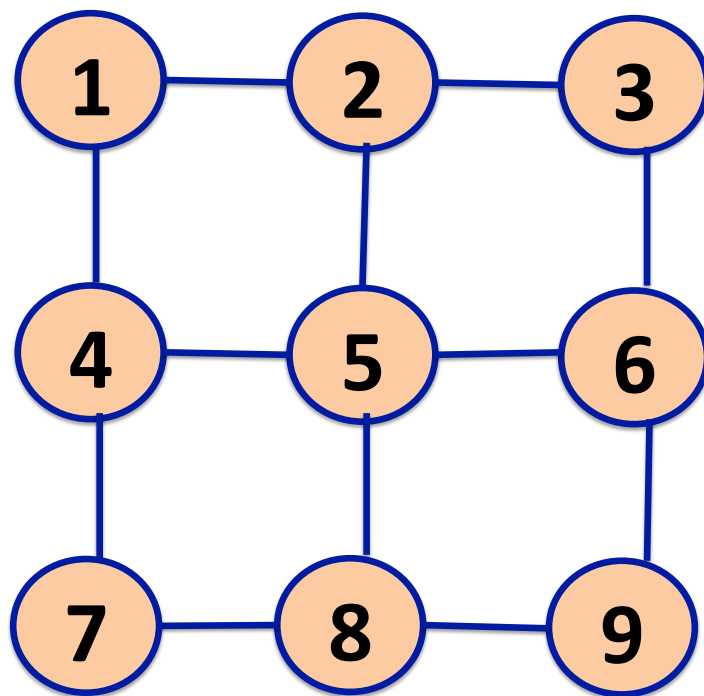
Maximum delay between the controller  
and any packet switch: 4

Worst case delay:

1, 3, 7 or 9 (Controller at a corner): 4

4, 6, 2 or 8: 3

5: 2



Scratch pad for p. 10

Completion time = cycles you need / speed

$$= 10^7 / 10^3 = 10^4 \text{ seconds}$$

Cost = completion time \* unit price

$$= 10^4 * 0.1$$

$$= 10^3$$

Scratch pad for p. 13

48% to Resource 1

Cycles needed from Resource 1

$$= 0.48 * 10^7 \text{ cycles}$$

=

Completion time

$$= 0.48 * 10^7 / 10^3$$

$$= 0.48 * 10^4$$

$$= 4.8 * 10^3$$

# Optimisation

Min/max an **objective** of many **decision variables**

Subject to:

**Constraints** on the decision variables

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**Feasible solution** : All constraints are satisfied

**Infeasible solution** : Not all constraints are satisfied

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Cloud resource allocation problem requirements:

Must be completed in at most 4800s.

Cost must not exceed 1500\$

Trial and error solution 2:  $T = 3500$  ,  $C = 1475$

Trial and error solution 3:  $T = 3000$  ,  $C = 1475$

■ Given:

- Job requirement =  $10^7$  million cycles
- Completion time  $\leq 4,800$  sec
- Budget  $\leq 1,500$  dollars

■ Let:

- $x_1$  = fraction of the job to Resource 1
- $x_2$  = fraction of the job to Resource 2
- $x_3$  = fraction of the job to Resource 3

■ Find  $x_1$ ,  $x_2$  and  $x_3$  such that

- All requirements are met
- Completion time is minimized

What is the maximum of 6, 10 and 7? 10

What is the minimum value of  $x$  such that  
 $x \geq 6$ ,  $x \geq 10$  and  $x \geq 7$ ? 10

## ■ Completion time:

■ Resource 1 =  $\frac{10^7 \times x_1}{1000} = 10000 \times x_1$

■ Resource 2 =  $\frac{10^7 \times x_2}{2000} = 5000 \times x_2$

■ Resource 3 =  $\frac{10^7 \times x_3}{3000} = \frac{10000}{3} \times x_3$

■ Job completion time

$$T = \max(10000 \times x_1, 5000 \times x_2, \frac{10000}{3} \times x_3)$$

$$\text{Cost } C = 1000 \times x_1 + 1250 \times x_2 + 2000 \times x_3$$

■ Let:

■  $x_1$  = fraction of the job to Resource 1

■  $x_2$  = fraction of the job to Resource 2

■  $x_3$  = fraction of the job to Resource 3

$$\min T$$

subject to

$$T \geq 10000 \times x_1$$

$$T \geq 5000 \times x_2$$

$$T \geq \frac{10000}{3} \times x_3$$

$$T \leq 4800$$

$$1000 \times x_1 + 1250 \times x_2 + 2000 \times x_3 \leq 1500$$

$$x_1 + x_2 + x_3 = 1$$

$$x_1, x_2, x_3 \geq 0$$

Any number in the interval  $[8,20]$

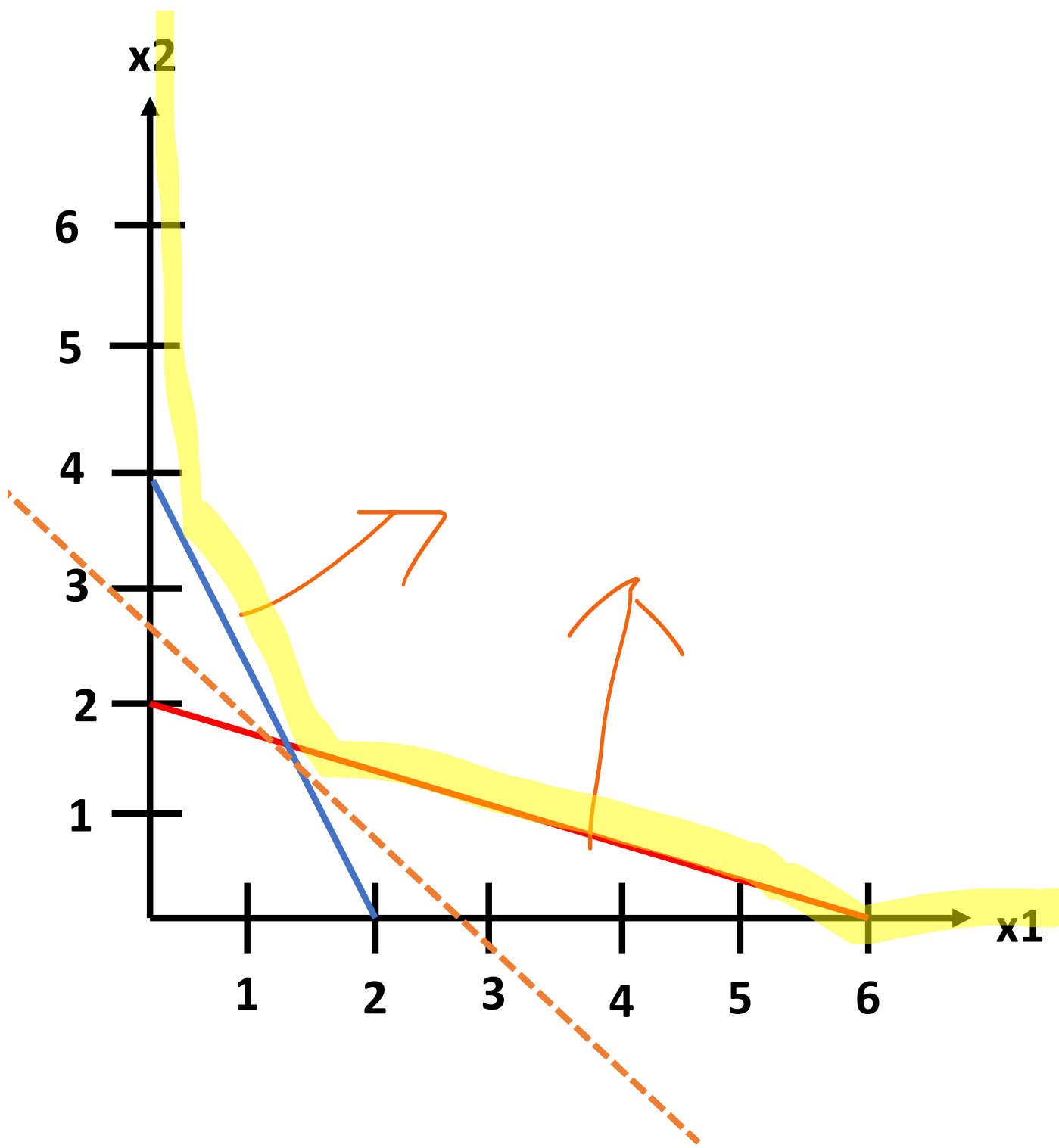
Optimal  $x = 8$

LP is short for linear programming

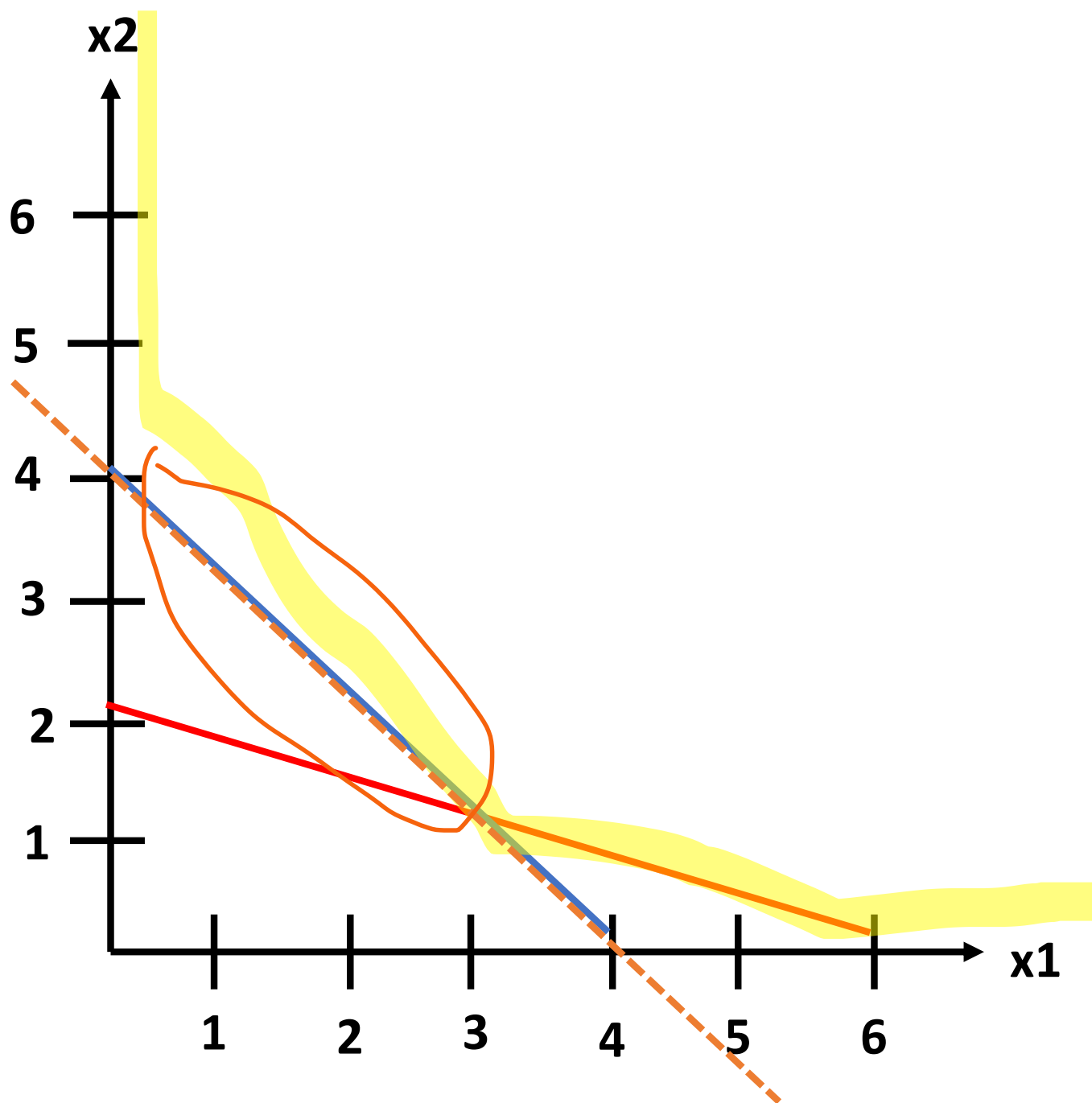
Objective and the constraints is linear  
in the decision variables



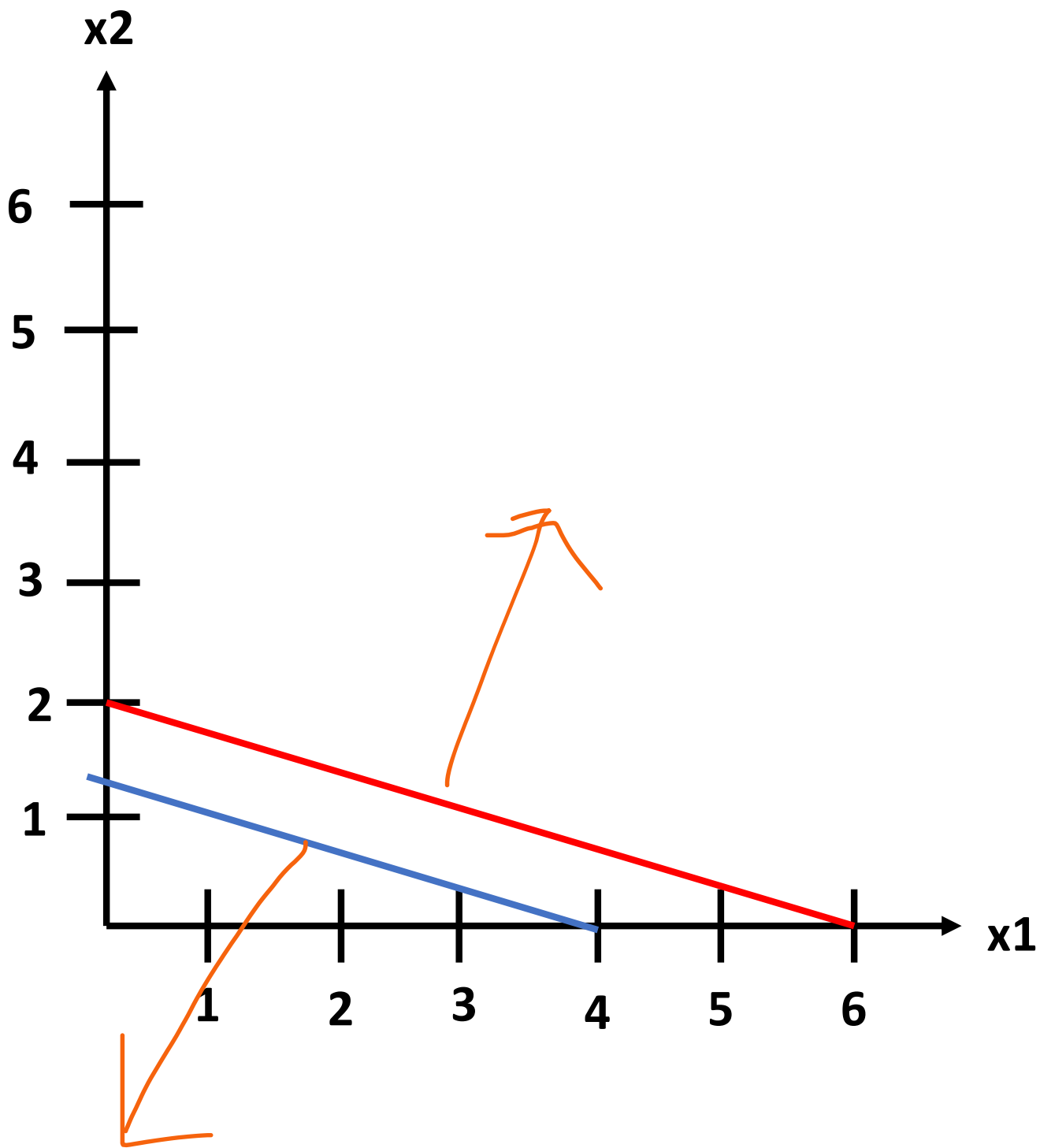
$$\begin{array}{ll}\min & x_1 + x_2 \\ \text{subject to} & 2x_1 + x_2 \geq 4 \\ & x_1 + 3x_2 \geq 6 \\ & x_1, x_2 \geq 0\end{array}$$



—

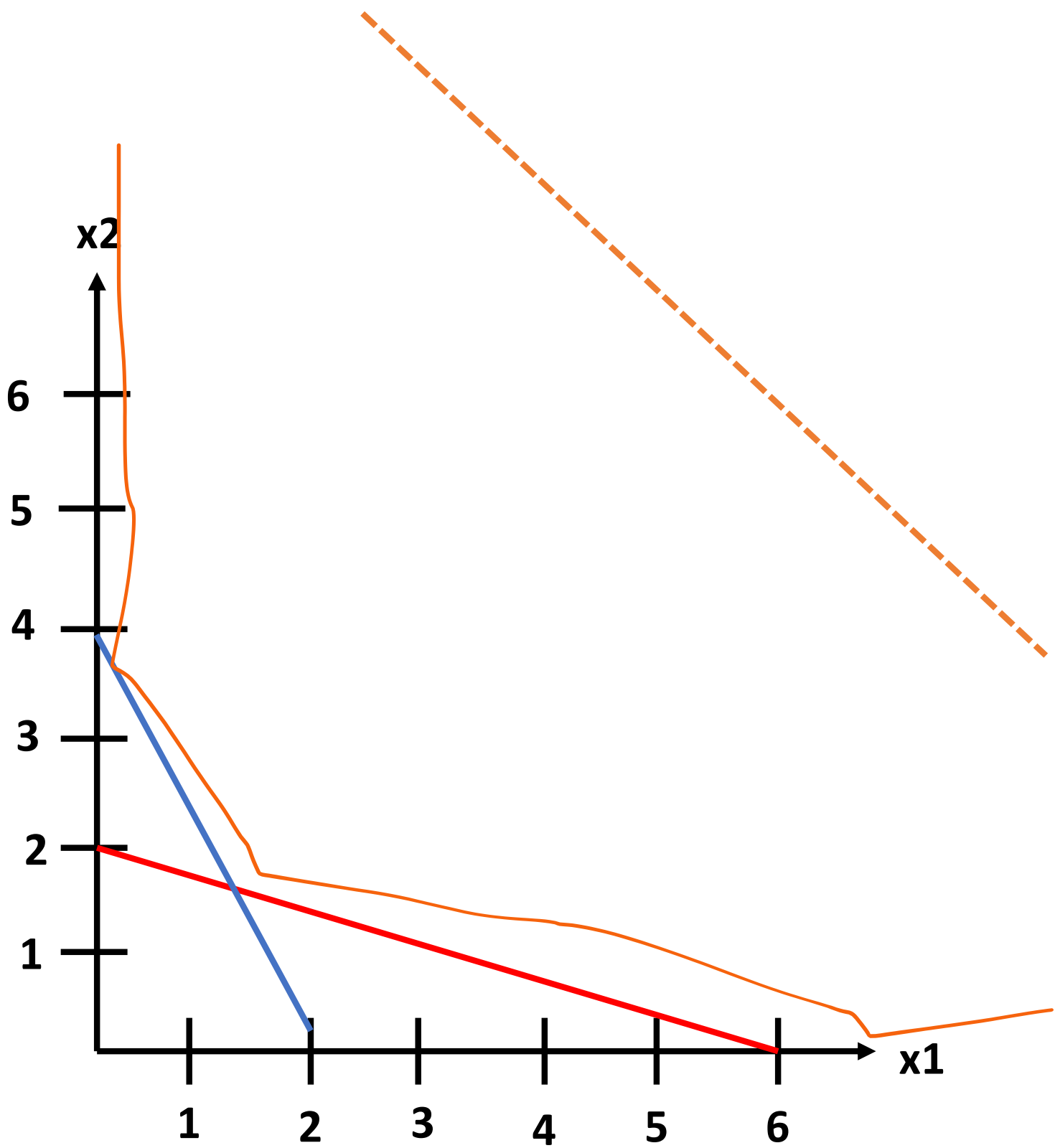


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■ Let:

- $x_1$  = fraction of the job to Resource 1
- $x_2$  = fraction of the job to Resource 2
- $x_3$  = fraction of the job to Resource 3

■ Completion time:

- Resource 1 =  $\frac{10^7 \times x_1}{1000} = 10000 \times x_1$
- Resource 2 =  $\frac{10^7 \times x_2}{2000} = 5000 \times x_2$
- Resource 3 =  $\frac{10^7 \times x_3}{3000} = \frac{10000}{3} \times x_3$
- Job completion time

$$T = \max(10000 \times x_1, 5000 \times x_2, \frac{10000}{3} \times x_3)$$

$$\text{Cost } C = 1000 \times x_1 + 1250 \times x_2 + 2000 \times x_3$$

■ Completion time:

■ Resource 1 =  $\frac{10^7 \times x_1}{1000} = 10000 \times x_1$

■ Resource 2 =  $\frac{10^7 \times x_2}{2000} = 5000 \times x_2$

■ Resource 3 =  $\frac{10^7 \times x_3}{3000} = \frac{10000}{3} \times x_3$

■ Job completion time

$$T = \max(10000 \times x_1, 5000 \times x_2, \frac{10000}{3} \times x_3)$$

$$\text{Cost } C = 1000 \times x_1 + 1250 \times x_2 + 2000 \times x_3$$

$$\min T$$

subject to

$$T \geq 10000 \times x_1$$

$$T \geq 5000 \times x_2$$

$$T \geq \frac{10000}{3} \times x_3$$

$$T \leq 4800$$

$$1000 \times x_1 + 1250 \times x_2 + 2000 \times x_3 \leq 1500$$

$$x_1 + x_2 + x_3 = 1$$

$$x_1, x_2, x_3 \geq 0$$



### Resource 3

Speed: 3,000 million  
cycles/sec

Cost: 0.6 dollars/sec

$p_3$

$N$

$c_3$

Completion time: Resource 3 =  $\frac{10^7 \times x_3}{3000} = \frac{10000}{3} \times x_3$

$$\text{Cost } C = 1000 \times x_1 + 1250 \times x_2 + 2000 \times x_3$$

■ E.g. is Resource 3 chosen?

■ Yes  $\Rightarrow x_3 > 0 \Rightarrow \text{Cost} = 2000 \times x_3 + 50 \times 1$

■ No  $\Rightarrow x_3 = 0 \Rightarrow \text{Cost} = 2000 \times x_3 + 50 \times 0$

$s_3$

Cost of using Resource 3 =

$$\frac{N x_3 c_3}{p_3} + s_3 y_3$$

Constraint:  $x_i \leq y_i$

In conjunction with  $x_i \geq 0$

If  $y_i = 0$ ,  $x_i \leq y_i$  means  
 $x_i \leq 0$

The two constraints on  $x_i$   
means  $x_i = 0$

If  $y_i = 1$ ,

$x_i \leq 1$  and  $x_i \geq 0$

$0 \leq x_i \leq 1$

$$\min \quad T$$

subject to

$$T \geq \frac{Nx_i}{p_i}, \quad i = 1, 2, \dots, n$$

$$T \leq T_{\max}$$

$$\sum_{i=1}^n \left( \frac{c_i N x_i}{p_i} + s_i y_i \right) \leq C_{\max}$$

$$x_i \leq y_i, \quad i = 1, 2, \dots, n$$

$$\sum_{i=1}^n x_i = 1$$

$$x_i \geq 0, \quad i = 1, 2, \dots, n$$

$$y_i \in \{0, 1\}, \quad i = 1, 2, \dots, n$$

```
set COMP := COMP_1 COMP_2 COMP_3;
```

| param: | p    | c    | s := |
|--------|------|------|------|
| COMP_1 | 1000 | 0.1  | 200  |
| COMP_2 | 2000 | 0.25 | 100  |
| COMP_3 | 3000 | 0.6  | 50;  |

```
param Tmax := 4800;
```

```
param Cmax := 1500;
```

```
param N := 1e7;
```

■ Given:

- $n$  resources
- Resource  $i$  offers computing power at a speed of  $p_i$  million cycles/sec with cost  $c_i$  dollars/sec
- Set up cost for using Resource  $i$  is  $s_i$  dollars
- Customer requires  $N$  million cycles
- Completion time  $\leq T_{\max}$
- Cost  $\leq C_{\max}$

■ Decision variables:

$$y_i = \begin{cases} 1 & \text{if Resource } i \text{ is chosen} \\ 0 & \text{otherwise} \end{cases}$$

$$x_i = \text{fraction of the job allocated to Resource } i$$

$$T = \text{completion time, which is } \max_i \frac{Nx_i}{p_i}$$

The constraint  $x_i \leq y_i$  does not prevent

$y_i = 1$  and  $x_i = 0$

from happening

but our optimization problem is still useful, why?

Cost of using Resource 3 =

$$\frac{N \ x_3 \ c_3}{p_3}$$

$$+ \ s_3 \ y_3$$

$$y_3 \ ($$

$$\frac{N \ x_3 \ c_3}{p_3}$$

$$+ \ s_3 \ )$$